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Norm of a matrix

Aim

To write a program to find the 1-norm, 2-norm and infinity norm of the matrix and display the result in two decimal places.

Equipment's required:

1. Hardware – PCs
2. Anaconda – Python 3.7 Installation / Moodle-Code Runner

Algorithm:

```
1. Get the input matrix using np.array()
2. Find the 2-norm of the matrix using np.linalg.norm()
```

3. Print the norm of the matrix in two decimal places.

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```
# Register No:212225230137
# Developed By:keerthana k
# 1-Norm of a Matrix
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
one_matrix=np.linalg.norm(matrix,1)
print("{:.2f}".format(one_matrix))
```

```
# 2-Norm of a Matrix

import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
two_matrix=np.linalg.norm(matrix,2)
print("{:.2f}".format(two_matrix))
```

```
# Infinity Norm of a Matrix
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
inf_matrix=np.linalg.norm(matrix,np.inf)
print("{:.2f}".format(inf_matrix))
```

Output:

1-Norm of a Matrix

For example:

Input	Result
[[-1, 3],[3, -4],[1, 7]]	14.00

Answer: (penalty regime: 0 %)

```
'''
To find the 1-Norm of a matrix and display the results in two decimal places.
Developed by : KEERTHANA K
Register no:212225230137
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
one_matrix=np.linalg.norm(matrix,1)
print("{:.2f}".format(one_matrix))
```

	Input	Expected	Got	
✓	[[-1, 3],[3, -4],[1, 7]]	14.00	14.00	✓
✓	[[1, 2, 3],[-3,-4,-1],[9,6,1]]	13.00	13.00	✓

Passed all tests! ✓

2-Norm of a Matrix

Input	Result
[[1,2],[3,4]]	5.46
[[-1, 3],[3, -4],[1, 7]]	8.66

Answer: (penalty regime: 0. %)

Reset answer

```
'''
Program to find 2-norm of a matrix.
Developed by: KEERTHANA K
RegisterNumber: 212225230137
'''
import os
os.environ["OPENBLAS_NUM_THREADS"]="1"
import numpy as np
matrix=eval(input())
two_matrix=np.linalg.norm(matrix,2)
print("{:.2f}".format(two_matrix))
```

	Input	Expected	Got	
✓	[[1,2],[3,4]]	5.46	5.46	✓
✓	[[-1, 3],[3, -4],[1, 7]]	8.66	8.66	✓
✓	[[2, 3],[3, 4],[1, 8]]	9.86	9.86	✓

Passed all tests! ✓

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